Chapter 15 — Basic Separation Techniques

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 324–325

Q.1 — Multiple Choice Questions (with correct option and reason)

I. After crystals form, they are separated from the mother liquor by:

(a) Evaporation
(b) Decantation or filtration
(c) Sublimation
(d) Distillation

Answer: (b) Decantation or filtration — Crystals are separated by filtration or decantation.

II. The comparative rates at which solutes move in paper chromatography depend on:

(a) Size of filter paper
(b) Rf values of solutes
(c) Temperature
(d) Size of the chromatographic jar

Answer: (b) Rf values of solutes — Different Rf values reflect different partitioning between phases.

III. The method used to separate two solids with different solubilities in a solvent is:

(a) Distillation
(b) Isolation
(c) Crystallization
(d) Filtration

Answer: (c) Crystallization — Crystallization exploits solubility differences.

IV. To check inks on a suspected altered legal document, use:

(a) Distillation
(b) Chromatography
(c) Solvent extraction
(d) Crystallization

Answer: (b) Chromatography — Chromatography separates and compares ink components.

V. In chromatography, components are separated based on:

(a) Their molecular masses
(b) Interaction with stationary and mobile phases
(c) The type of solvent used
(d) The filter paper used

Answer: (b) Interaction with stationary and mobile phases — Separation depends on differential distribution between the two phases.

VI. The porous material used to separate solid from liquid in filtration is the:

(a) Filtrate
(b) Residue
(c) Filter medium
(d) Solvent

Answer: (c) Filter medium — The filter medium (e.g. filter paper) retains the residue.

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VII. The key difference between simple and fractional distillation is the presence of a:

(a) Condenser with larger surface area **(b)** More powerful heat source
(c) Fractionating column **(d)** Vacuum pump

Answer: (c) Fractionating column — Fractional distillation uses a fractionating column.

VIII. For simple distillation, the boiling points should differ by at least:

(a) 5 °C **(b)** 10 °C
(c) 25 °C **(d)** 50 °C

Answer: (c) 25 °C — The textbook states a difference of around 25 °C is needed for effective separation.

Q.2 — Short Answer Questions

a. Why is there a need to crystallize a crude product?

Crystallization purifies a crude solid by removing soluble and insoluble impurities, giving pure crystals.

b. What is the function of fluted filter paper?

Fluting increases the surface area and speeds filtration by allowing air to escape and liquid to pass faster.

c. What is the difference between a Gooch crucible and a sintered glass crucible?

A Gooch crucible has a perforated base used with an asbestos/filter mat; a sintered glass crucible has a porous glass disc and needs no mat.

d. What type of mixtures are filtered through a Gooch crucible?

Mixtures containing fine or gelatinous precipitates that might clog ordinary filter paper, especially when the solid must be ignited/weighed.

e. What is the function of the fractionating column in fractional distillation?

It provides repeated vaporization–condensation cycles, enriching the vapour in the more volatile component and improving separation.

f. What is the stationary phase in paper chromatography?

Water held (adsorbed) on the cellulose fibres of the paper.

g. What happens if components have comparable attraction for the stationary phase?

They move at similar rates and do not separate well (spots overlap).

h. What is meant by “developing the chromatogram”?

Allowing the mobile phase (solvent) to move up the paper by capillary action so the components separate into spots.

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i. What is the basic principle of paper chromatography?

Separation based on differential partitioning of components between a stationary phase (water on paper) and a mobile phase (solvent).

j. Why is water not generally used as a solvent in paper chromatography?

Water is already the stationary phase; using it as mobile phase gives poor separation because components do not partition differently.

k. Differentiate between adsorption and partition chromatography.

Adsorption: separation by differential adsorption on a solid stationary phase. Partition: separation by differential solubility/partition between two liquid phases (stationary liquid on a support and mobile liquid).

l. How can you check purity with paper chromatography?

A pure substance gives a single spot; more than one spot indicates impurities.

m. You prepared solid glucosazone. How will you check its purity?

Determine its melting point (pure = sharp m.p.) and/or run paper/TLC chromatography against a standard; a single spot and matching Rf/m.p. indicate purity.

Descriptive / Long Questions

Q.3 Differentiate simple and fractional distillation in construction and applications.

Feature	Simple distillation	Fractional distillation
Construction	Distillation flask + condenser, no column	Adds a fractionating column (glass beads)
Boiling-point difference	Large (≳25 °C)	Small/close boiling points
Separation	One vaporization–condensation	Repeated vaporization–condensation
Application	Solvent from a solution (e.g. pure water from seawater)	Miscible liquids (e.g. ethanol–water, petroleum fractions)

Q.4 Describe the criterion to choose a suitable solvent for crystallization.

The solvent should dissolve the solute well when hot but sparingly when cold; it should not react with the solute; impurities should either remain soluble when cold or be insoluble when hot (so they can be filtered); and it should be easily removable (volatile) and safe.

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Q.5 What is the basic principle of paper chromatography? Give three practical life applications.

Principle: components partition differently between a stationary phase (water on paper) and a mobile phase (solvent), so they travel at different rates and separate.

Applications: (1) separating/identifying pigments and dyes; (2) checking purity of compounds/pharmaceuticals; (3) forensic analysis of inks; (4) separating amino acids/sugars in biochemistry.

Quick Check Solutions (in-chapter practice boxes)

Quick Check 15.1 (book p. 317)

a. Name different filter media used in filtration.

Filter paper (flat/fluted), glass wool, cotton, sintered glass, Gooch crucible with asbestos mat, and membrane filters.

b. Components of the filtration process.

The mixture, the filter medium, the residue (solid retained) and the filtrate (liquid that passes through).

c. Differentiate gravity filtration and vacuum filtration.

Gravity filtration relies on gravity to pull liquid through the filter (slower, used for hot solutions). Vacuum filtration uses reduced pressure to speed filtration (used for collecting crystals).

d. Differentiate Gooch and sintered glass crucible.

A Gooch crucible has a perforated base and uses an asbestos/filter mat; a sintered glass crucible has a porous glass disc and needs no mat.

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Quick Check 15.2 (book p. 319)

a. Basic principle of crystallization.

A solid is dissolved in a suitable hot solvent and then recovered as pure crystals on cooling, because solubility decreases with temperature and impurities remain in solution.

b. How a suitable solvent is selected.

It should dissolve the solute well when hot but sparingly when cold, not react with the solute, allow impurities to remain soluble or be filtered off, and be easily removable/safe.

c. Important steps of crystallization.

(1) Choose solvent; (2) dissolve crude solid in minimum hot solvent; (3) hot filtration to remove insoluble impurities; (4) cool slowly to crystallize; (5) filter crystals; (6) wash and dry.

Quick Check 15.3 (book p. 320)

a. Difference between simple and fractional distillation.

Simple distillation separates a liquid from a dissolved solid or liquids with very different boiling points (one vaporization–condensation). Fractional distillation uses a fractionating column for repeated vaporization–condensation to separate liquids with close boiling points.

b. Daily-life applications of fractional distillation.

Petroleum refining (petrol, diesel, kerosene fractions), separation of ethanol–water, and production of alcoholic spirits/industrial solvents.

Quick Check 15.4 (book p. 323)

a. On what basis are components separated in paper chromatography?

On their differential partitioning between the stationary phase (water on paper) and the mobile phase (solvent), giving different rates of movement.

b. Stationary and mobile phases in paper chromatography.

Stationary phase: water held on the cellulose fibres of the paper. Mobile phase: the solvent that moves up the paper by capillary action.

c. How purity is determined by paper chromatography.

A pure substance gives a single spot; more than one spot indicates impurities. Matching Rf with a standard also supports identity/purity.

Quick Check 15.5 (book p. 323)

a. How melting point indicates purity.

A pure solid has a sharp, definite melting point; an impure solid melts over a lower and broader temperature range.

b. Suitable technique for: (i) petroleum and water (ii) NaCl and $NaNO_3$ (iii) activated charcoal and sodium chloride (iv) methanol, ethanol and propanol.

(i) Separating funnel/decantation (immiscible liquids). (ii) Fractional crystallization (different solubilities). (iii) Filtration (charcoal insoluble) followed by crystallization of NaCl. (iv) Fractional distillation (different boiling points).

Quick Check 15.3(b): the book prints “fractional filtration”; the intended term is fractional distillation.

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